

Traffic Light System using Arduino Uno

Abstract

This project demonstrates a basic traffic light controller designed and simulated using Tinkercad Circuits. Three LEDs represent red, yellow, and green traffic signals. Buzzers provide audible indication for each signal. The Arduino Uno controls the sequence using timed delays.

Objectives

- Learn Arduino digital output programming.
- Understand sequential control logic.
- Simulate a traffic signal using LEDs and buzzers.
- Practice circuit design using Tinkercad.

Components Required

Arduino Uno, Breadboard, Red LED, Yellow LED, Green LED, 3 × 220Ω Resistors, 3 Buzzers, Jumper Wires.

Software Used

Tinkercad Circuits, Arduino IDE, Embedded C/C++.

Working Principle

The Arduino switches ON the LEDs in the order Red → Yellow → Green → Yellow. Each state is maintained using delay() to simulate a real traffic signal. A corresponding buzzer sounds whenever its LED is active.

Pin Configuration

Green LED → D8
Yellow LED → D12
Red LED → D13

Applications

Traffic signal simulation, Embedded systems education, Digital electronics laboratory, Arduino programming practice.

Learning Outcomes

Digital output interfacing, LED control, Timing functions, Embedded C programming, Circuit simulation.

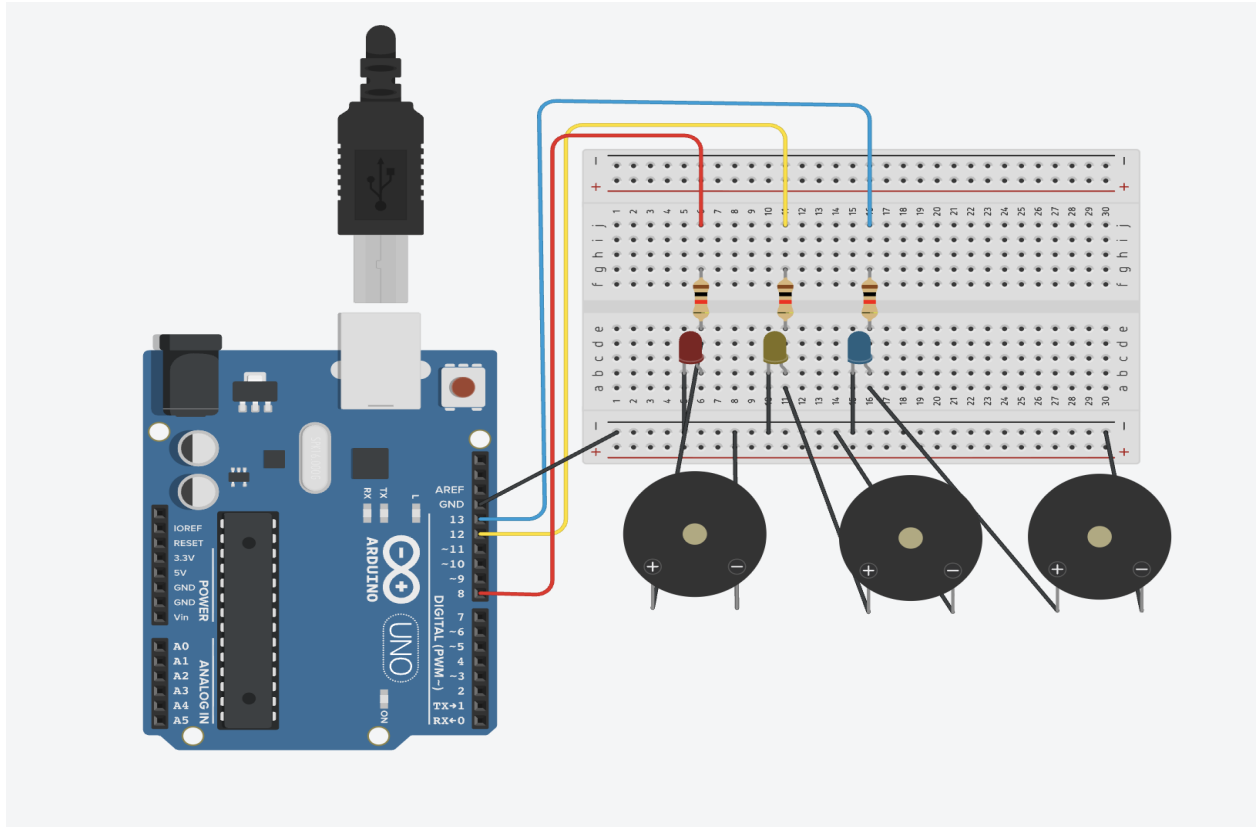
Conclusion

The project successfully simulates a simple traffic light sequence using Arduino Uno. It provides a practical understanding of embedded programming and digital output control.

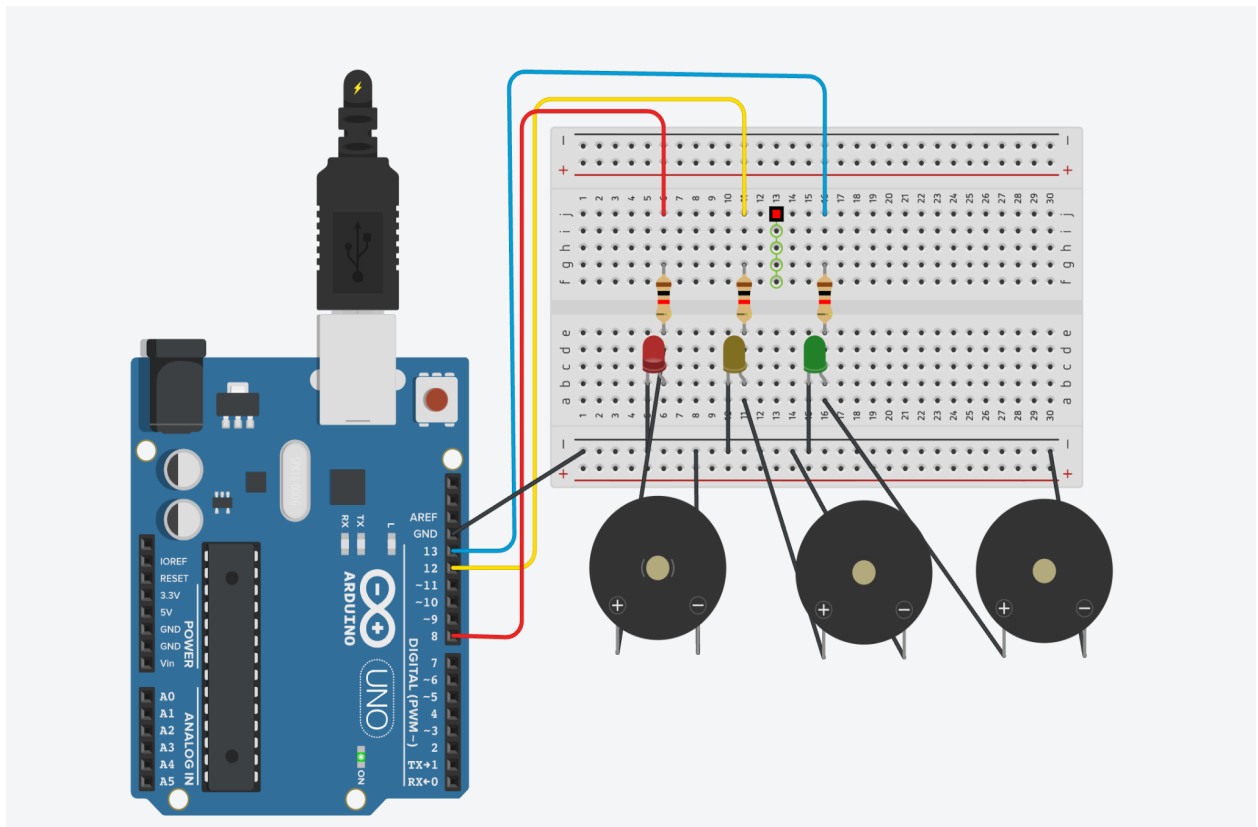
Arduino Source Code

```
1 // Traffic Light System
2
3 // Define LED pins
4 int greenLED = 8;
5 int yellowLED = 12;
6 int redLED = 13;
7
8 void setup() {
9     pinMode(greenLED, OUTPUT);
10    pinMode(yellowLED, OUTPUT);
11    pinMode(redLED, OUTPUT);
12 }
13
14 void loop() {
15
16     // RED Light - STOP
17     digitalWrite(redLED, HIGH);
18     digitalWrite(yellowLED, LOW);
19     digitalWrite(greenLED, LOW);
20     delay(5000);
21
22     // YELLOW Light - GET READY
23     digitalWrite(redLED, LOW);
24     digitalWrite(yellowLED, HIGH);
25     digitalWrite(greenLED, LOW);
26     delay(2000);
27
28     // GREEN Light - GO
29     digitalWrite(redLED, LOW);
30     digitalWrite(yellowLED, LOW);
31     digitalWrite(greenLED, HIGH);
32     delay(5000);
33
34     // YELLOW Light - SLOW DOWN
35     digitalWrite(redLED, LOW);
36     digitalWrite(yellowLED, HIGH);
37     digitalWrite(greenLED, LOW);
38     delay(2000);
39 }
```

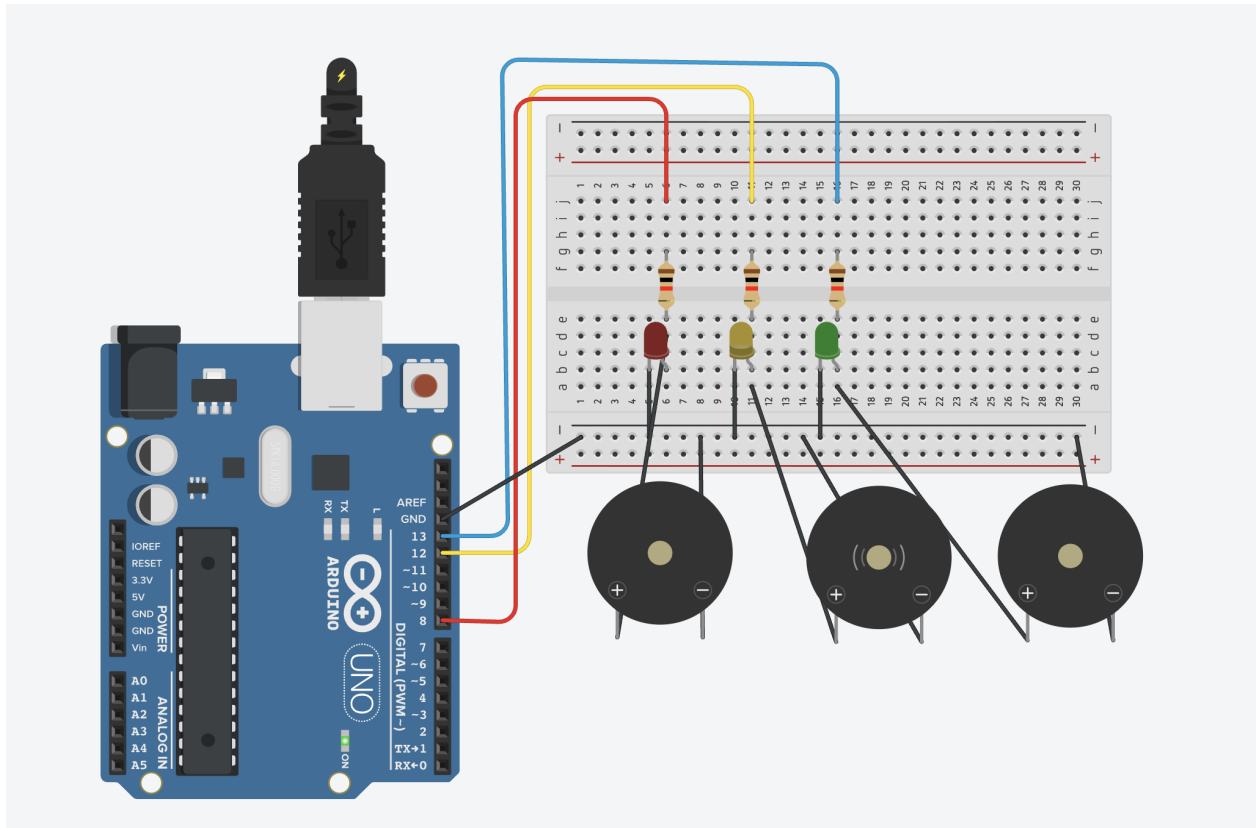
Circuit Diagram



Output - Red Signal



Output - Green Signal



Output - Yellow Signal

